

Ruchi – Architecture

Agents for Humans · Professional Agents track

A citation-integrity agent for Indian case law. It answers a lawyer's question by running deterministic checks over the entire Supreme Court of India, and it is forbidden from asserting anything a check did not return.

Corpus	38,032 judgments · 38,005 with full text · 707,647 paragraphs · 14 March 1950 – 12 December 2025 · Supreme Court of India only
Data	AWS Open Data, Indian Supreme Court Judgments, CC-BY-4.0
Agent	Strands Agents SDK — one agent, eight tools, one per check
Model	Amazon Bedrock by default; provider chain via LiteLLM as fallback, reported in every answer
Engine	LangGraph state graph, 8 nodes, fixed order · BM25 + FTS5 · SQLite
Serving	FastAPI + Typer CLI · Docker on EC2 · Apache-2.0

1. The system

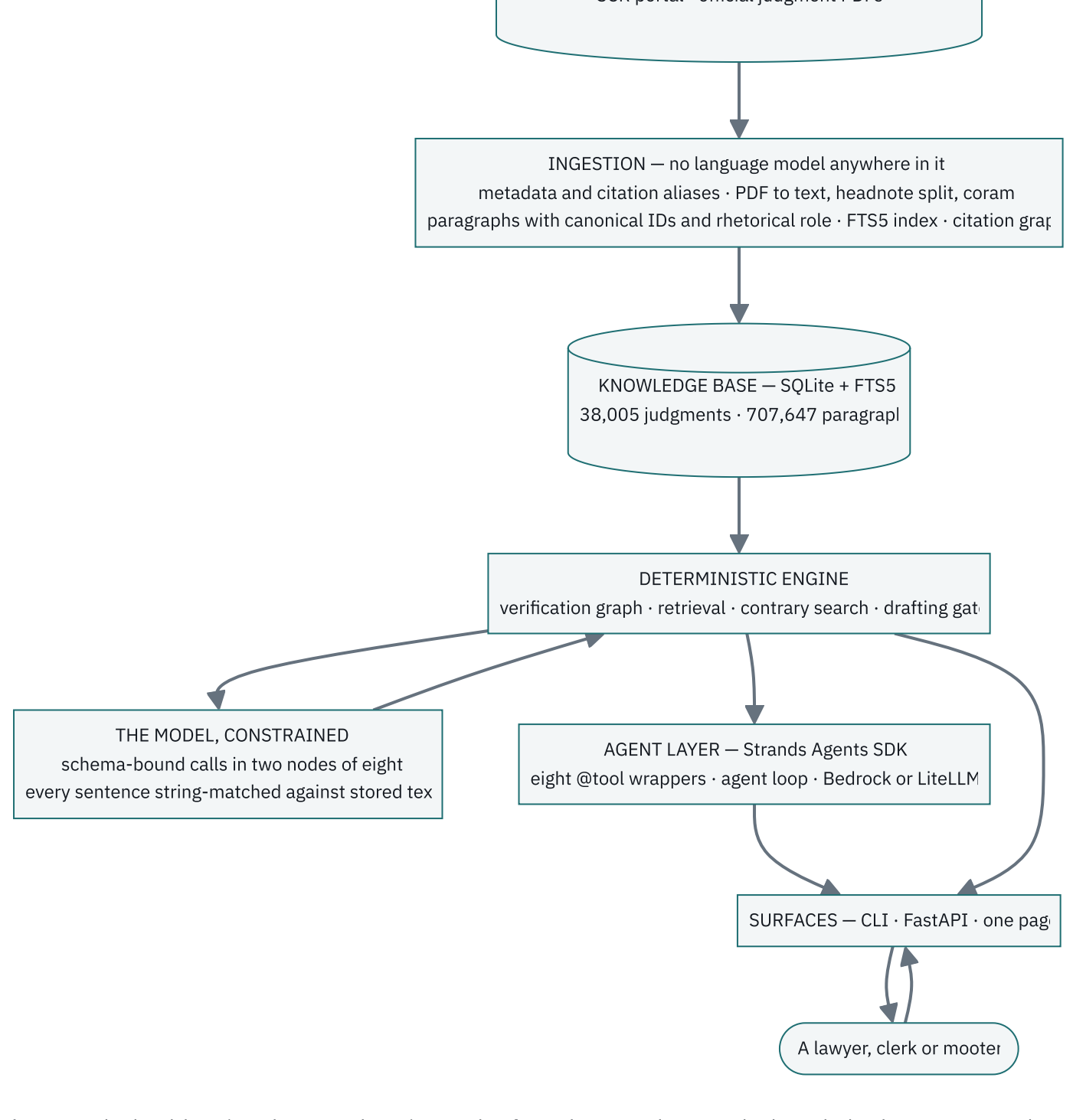


Figure 1. The load-bearing absence: there is no edge from the agent layer to the knowledge base. Every path runs through a tool, and every tool through a check that existed before the agent did. The agent chooses which question is asked; it never decides the answer.

The engine was built first and the agent last. That order is why the agent could be added without relaxing anything: each tool is a wrapper around an entry point that already had a test, a CLI command and a caller.

2. The agent layer

Built with the Strands Agents SDK. Strands runs the loop — question in, tool calls out, results back, answer — and the eight tools are the checks the engine already performed. Each tool opens a database session, calls one existing function, and returns JSON.

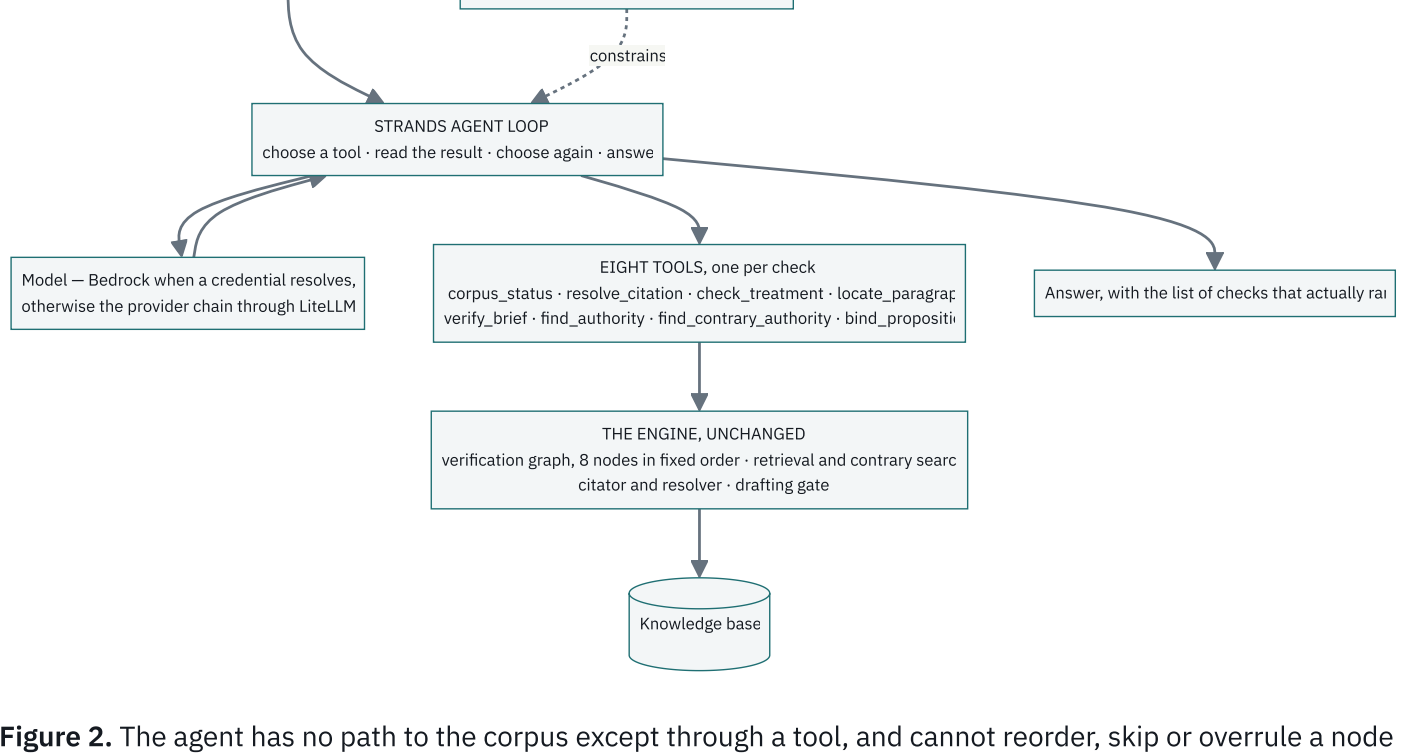


Figure 2. The agent has no path to the corpus except through a tool, and cannot reorder, skip or overrule a node of the verification graph.

Tool	Wraps	Answers
corpus_status	row counts, index state	what is held, so "not in this corpus" is distinguishable from "no such authority"
resolve_citation	resolver	does this case exist, and which judgment does the citation name
check_treatment	citator	is it still good law, and what did later benches do with it
locate_paragraph	locator	which paragraph carries the claim, and does the cited one
verify_brief	verification graph	every citation in a passage, all eight stages, findings by mode number
find_authority	retrieval	which judgment backs a proposition, and which line
find_contrary_authority	contrary search	which judgment says the opposite
bind_proposition	drafting gate	may this sentence be written, and behind which authority

Why the trail matters. Every answer returns the tools that actually ran. An answer about subsequent history that never called `check_treatment` came out of the model's memory, which the prompt forbids — and that is visible without reading a log.

3. The verification engine

A LangGraph state graph. Each stage is a node, the state is the verdict in progress, and the path a citation takes is fixed rather than chosen — which is what makes a verdict auditable rather than merely produced. A language model is called in two nodes only.

Three states, never collapsed. Every claim is **supported**, **checked and not supported**, or **not checked**. A provider outage, a missing key or a judgment held as metadata only produces the third — never the first or second. A treatment of `good_law` carries unchecked: `true` when nothing in the corpus has ever cited that judgment: good law by default, not by evidence.

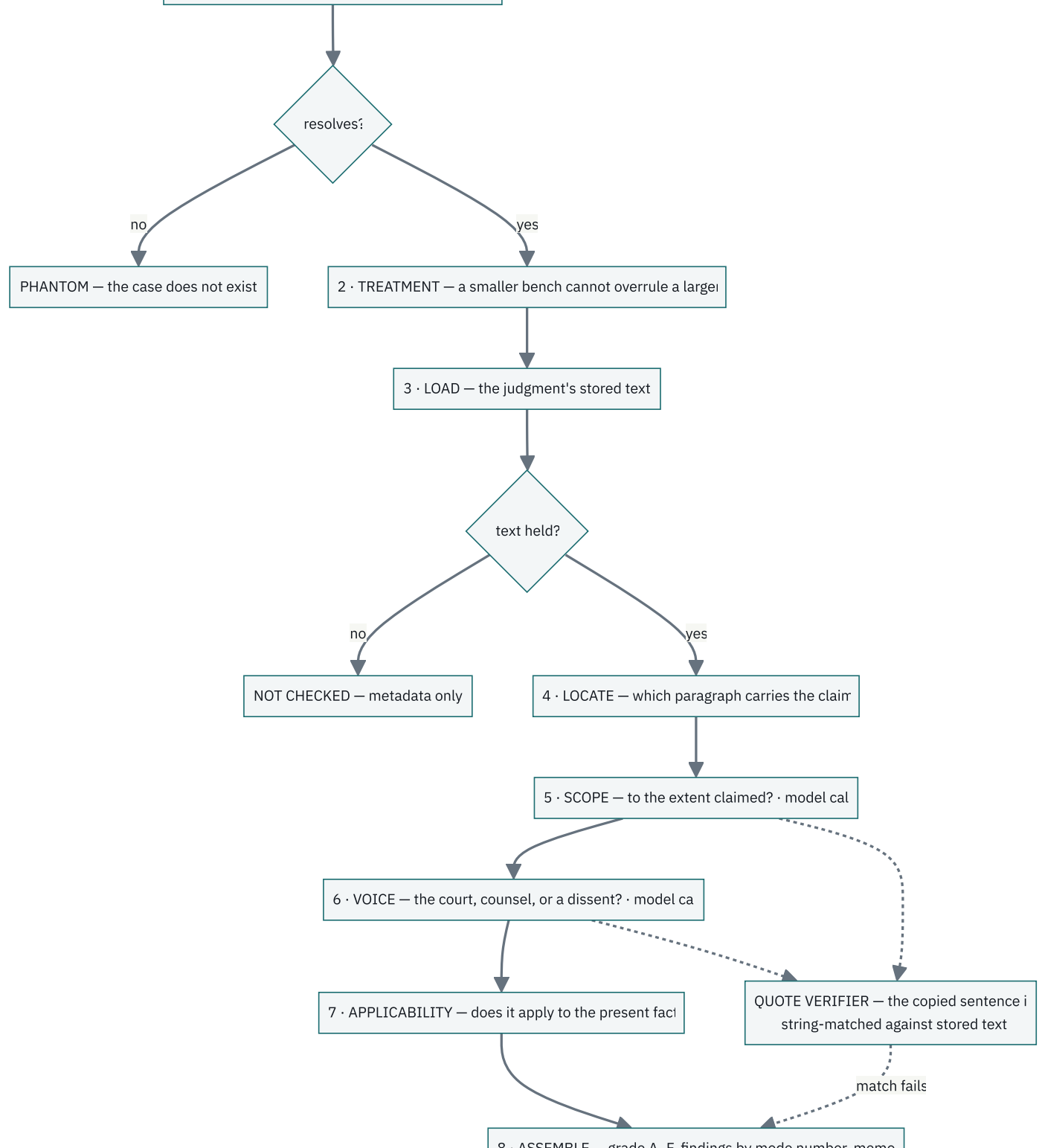


Figure 3. Eight of the twelve failure modes are decided before any model is called. A failed quote match downgrades the claim to unsupported regardless of the model's confidence.

4. Retrieval, in both directions

The same field of paragraphs is read twice: once for the line that supports a proposition, once for the line that contradicts it. What separates this from a text search is the voice filter — an advocate states a rule more baldly than a judge ever will, so counsel's submission out-matches the actual holding on keywords.

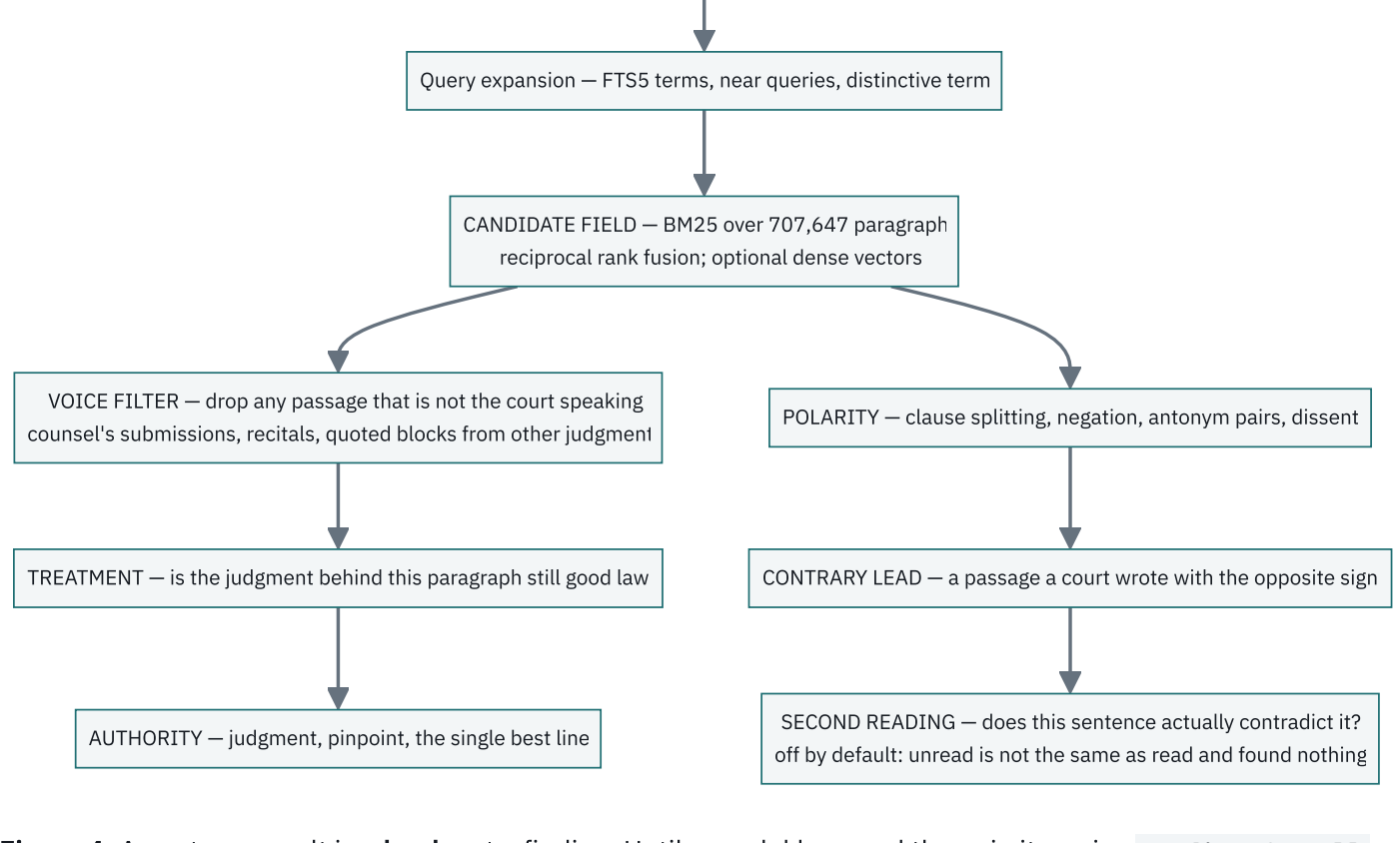


Figure 4. A contrary result is a **lead**, not a finding. Until a model has read the pair, it carries `confirmed: null` — and null is not false.

5. The drafting gate

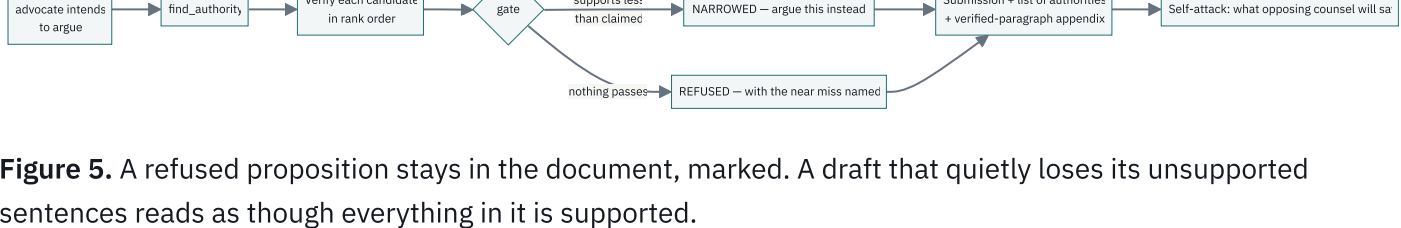


Figure 5. A refused proposition stays in the document, marked. A draft that quietly loses its unsupported sentences reads as though everything in it is supported.

6. Deployment

One EC2 instance, one EBS volume, one container. The corpus is state that outlives any version of the code, so it is a volume rather than an image layer; keys arrive as environment at run time, because a key in a layer is published to everyone who can pull the image.

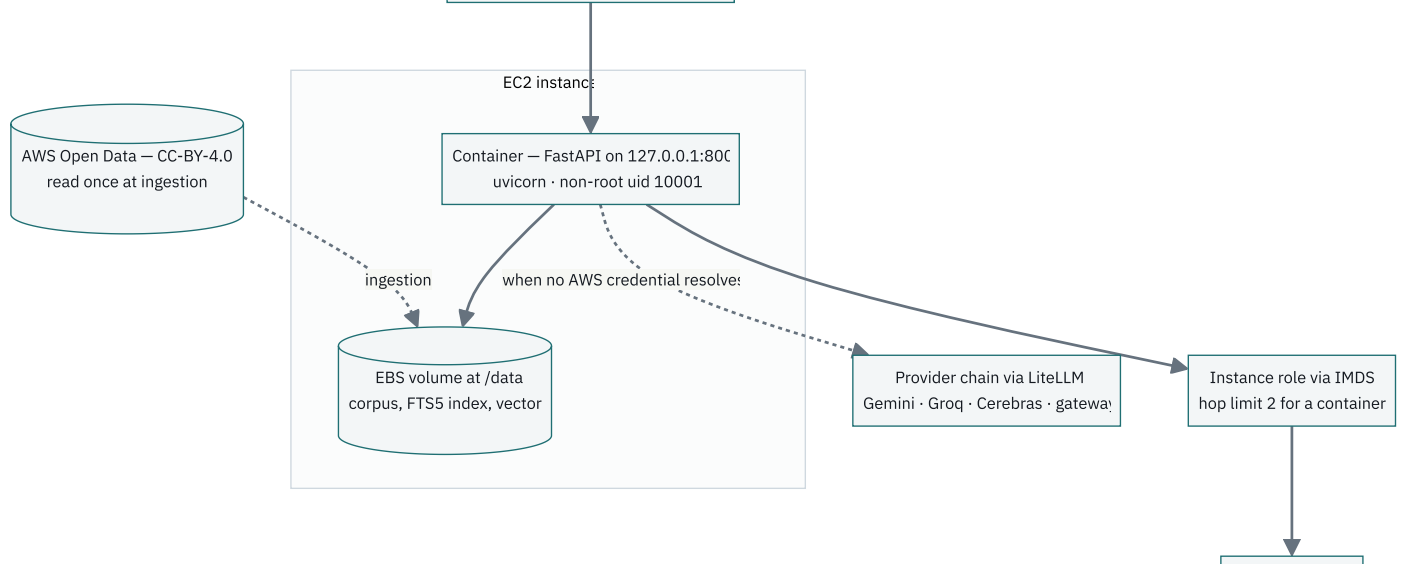


Figure 6. Deployed by GitHub Actions on a push to `production`: rebuild, restart, keep the volume. The token has no default and the server refuses to start off-loopback without one.

The failure that looks like success. A container cannot reach the instance role unless the IMDS hop limit is raised to 2. When the credential does not resolve, the agent falls back to the provider chain and keeps answering — so every answer, `/api/health`, and `orderorder agent --which report which model actually replied`.

What it scores

Measured on **forty judgments the detectors were not developed against**, against ground truth the corpus supplies rather than labels anyone wrote.

	Held-out set
no model	phantom 40/40 · mis-cite 40/40 · wrong bench 39/39 · wrong voice 34/34 · selective quotation 20/20 · dead law 14/14 · wrong pinpoint 40/40 · false positives 0/40
with a model	obiter as ratio 3/3 · false positives 0/25 · quote grounding 100% · abstention 68% · ~6 s a citation